How to Write a Machinery Specification Document

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Most factory owners do not know how to write a specification document. They send the supplier a description, such as “I need a packaging machine for 500 units per hour”, and the supplier builds to their default specification, not the client’s actual requirement.

This guide explains how to write a specification document that eliminates ambiguity and prevents the compatibility failures that destroy capital. Use it with the 48-point Pre-Import Checklist: the checklist tells you what to ask, and this guide shows you how to write the answers into a document your supplier can be held to.

Who this is for: factory owners, operations managers and procurement officers who are about to request quotations for machinery from China.

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1. The difference between a description and a specification

A **description** says what you want. A **specification** says exactly what must be delivered, in terms that can be measured, written into the contract, and checked at inspection.

Description	Specification
Packaging machine, 500 units per hour	Horizontal flow-wrap machine, 500 units per hour sustained over an 8-hour shift, for 200 g bars at 120 mm length, 415V three-phase 50Hz
Suitable for Nigerian power	Rated for 415V ±10%, three-phase, 50Hz, with a stabiliser supplied if the tolerance is exceeded
English manual included	Operation manual, wiring diagram and HMI in English, supplied before shipment

The three tests. A requirement belongs in your specification only if it passes all three:

1. **Measurable.** It uses a number, a unit or a named standard.
2. **Contractual.** It is written into the quotation and the purchase order, not only in chat messages.

3. **Checkable.** An inspector or your team can verify it before the balance is paid.

Words like “high quality”, “suitable”, “standard” and “as required” fail all three. Replace them with figures.

2. How to specify voltage, phase, and frequency for Nigerian power conditions

This is the most common cause of failure, so do it first.

Step 1: Measure your supply. Do not rely on the supply you are supposed to have. Ask a qualified electrician to measure at the point where the machine will connect, under normal load, at different times of day. Record the voltage between phases, the frequency, and the lowest and highest readings you see.

Step 2: Know the usual gap. Nigerian industrial supply is typically 415V three-phase at 50Hz, with fluctuations that can reach ±15-20%. Many Chinese factories build to 380V unless told otherwise. A machine built for 380V and run on 415V can overheat its motor and damage its control panel. Your measurements, not these typical figures, go in the document.

Step 3: Write all five items.

Item	Example wording
Voltage	415V between phases
Phase	Three-phase
Frequency	50Hz
Tolerance	Machine must operate within ±15% of nominal voltage without fault; supplier to state in writing if a stabiliser is required

Power load	Total connected load in kW or kVA stated in the quotation

Step 4: Declare your power source. If the site runs on a generator for any part of the day, say so and state its rating. Ask the supplier to confirm the machine's start-up current and that it is compatible.

Sample clause:

> "The machine shall be manufactured for a supply of 415V three-phase, 50Hz, tolerance ±15%. All motors, drives, transformers and control components shall be rated accordingly. The motor nameplate figures shall match this specification and be photographed at pre-shipment inspection. Any deviation is grounds for rejection."

Common mistakes: accepting "380V is standard" as an answer, skipping the tolerance figure, and not telling the supplier about generator use.

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3. How to specify material input requirements

A machine built for one material can fail on another that looks similar. A machine built for maize cannot process cassava, and a packaging machine set for one film will jam on another.

Describe your material with figures:

- Exact name and variety (for example "fresh cassava roots", not "tubers")
- Moisture content, as a percentage
- Size and shape: length, diameter, particle size or thickness
- Hardness, density or grade, where relevant
- Temperature at feed
- Likely contaminants (soil, stones, metal fragments, fibre)
- Form it arrives in: bulk, bagged, rolls, sheets
- Feed rate you plan to use

Ask for a test. Offer to send a sample, or ask for a video of the supplier running your material. Do not accept a test on a different material.

Sample clause:

> “The machine shall process fresh, unpeeled cassava roots, 15–40 cm long, 4–10 cm diameter, 60–70% moisture, with up to 5% soil by weight. Supplier shall provide a recorded factory test using this material before shipment.”

Common mistakes: naming the product instead of the material, ignoring moisture, and assuming the supplier knows local raw materials.

4. How to specify output requirements (capacity, speed, quality)

“500 units per hour” means nothing until you say under what conditions, and what counts as a good unit.

Capacity and speed

- Unit of output (kg, units, litres, metres)
- Rate per hour, and whether it is peak or sustained
- Duration the rate must hold (for example, 8 hours continuous)
- The material and settings it applies to

Quality

- Tolerance (for example, ±0.05 mm, fill accuracy ±2 g, cut length ±1 mm)
- Surface finish, seal strength, particle size, or purity, as relevant
- Acceptable waste or reject rate

Acceptance and remedy. State how output will be proved: a factory run witnessed by the inspector or recorded on video, measured against your figures. Then state what happens if it falls short: correction, delay or refund.

Sample clause:

> “Guaranteed output: 500 kg/hour sustained for 8 hours, fill accuracy ±2%, reject rate below 1%. Verified by a witnessed run at the factory. If output is below 95% of this figure, the supplier shall correct the machine at its own cost before shipment.”

5. How to specify control interface language

Mandarin-only interfaces are common, and an operator who cannot read the panel cannot run the machine safely.

Specify English for all of these:

- Control panel and HMI screens
- Alarm and fault messages
- Buttons, labels and safety signs
- Operation and maintenance manuals
- Electrical wiring diagrams and parts lists

Also state:

- Units are metric
- The make and model of the PLC and drives (so you can check local support)
- You will have administrator access to settings, and any software or licence needed is supplied
- Screenshots of the interface will be sent before production

Sample clause:

> "All HMI screens, alarm messages, labels, manuals and wiring diagrams shall be in English, units in metric. Supplier shall send HMI screenshots for approval before production begins."

6. How to specify spare parts requirements

A machine with no spare parts supply is a disposable asset. Decide what you need before you pay, not after the first breakdown.

Ask for these in writing:

1. A list of wear parts (belts, bearings, seals, blades, filters, sensors, heating elements) with part numbers
2. The brand and model of the critical parts: PLC, drives, motors, pumps
3. A spare parts kit included in the price, with a contents list
4. A price list and lead time for each important part
5. Which parts can be bought locally
6. Who pays shipping on parts replaced under warranty

Part	Brand / model	Quantity in kit	Available locally?	Lead time from China
Example: main motor				

Sample clause:

> "Supplier shall include a spare parts kit as listed in Annex B. Brands and part numbers for the PLC, drives and motors shall be stated. Replacement parts shall be available for at least five years, and parts replaced under warranty shall be shipped at the supplier's cost."

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7. How to specify installation and commissioning requirements

Installation problems often appear only after the container arrives, when it is expensive to fix.

Give the supplier your site information: room dimensions, door and access sizes, floor type and loading, crane or forklift access, ceiling height, available utilities, and ambient conditions (heat, dust, humidity).

Ask the supplier for:

- Machine dimensions and weight
- Foundation or anchoring drawings
- A utility requirements sheet: electrical load, compressed air pressure and flow, water supply, cooling, extraction
- An installation manual in English

Settle who does what:

- Who installs and commissions, and whether it is in the price
- The commissioning test: what is run, with what material, and who must be present
- Operator training: where, how long, and in what language
- When the machine counts as accepted: for example, after a successful on-site test at the stated output

Sample clause:

> “Supplier shall provide foundation drawings and a utility requirements sheet within 14 days of order. Commissioning shall include a test run at the guaranteed output with the buyer’s material and operators present. Acceptance occurs when the test is passed and recorded.”

8. How to specify documentation requirements

Documents that do not match each other delay customs clearance, and every extra week at the port adds cost.

Required with every shipment:

- Commercial Invoice
- Packing List
- Bill of Lading
- Certificate of Origin
- Operation, maintenance and wiring documents in English
- The factory test report and the third-party inspection report

Consistency rules. The company names, machine description, quantity, value and HS code should match across the Form M, Commercial Invoice and Packing List. Agree the machine description wording with your clearing agent before the supplier issues any document.

Timing. Ask for draft documents before the container ships, so errors are fixed while the goods are still at the factory. Confirm any certification or import approval your machine needs with your clearing agent.

Sample clause:

> “Supplier shall send draft Commercial Invoice, Packing List and Certificate of Origin for buyer approval at least 7 days before shipment. Descriptions, values and HS code shall match the buyer’s Form M. Final documents shall not be released until the buyer confirms.”

9. Put it together: one-page specification template

Copy this, fill in your figures, and attach it to your request for quotation.

Section	Your requirement
Machine	Type, model, quantity
Power	Voltage / phase / frequency / tolerance / total load / generator use
Material input	Material, moisture, size, temperature, contaminants, feed rate
Output	Rate, duration, quality tolerance, reject rate, acceptance test
Interface	English HMI, alarms, manuals, wiring diagrams, metric units
Controls	PLC, drive and motor brands, administrator access
Spare parts	Kit contents, brands, availability, lead time, warranty shipping
Installation	Foundation drawings, utility sheet, installer, commissioning test, training
Inspection	Third-party firm, scope, report before balance payment
Documents	Invoice, packing list, bill of lading, certificate of origin, reports
Payment	Milestones, release conditions
Warranty	Duration, coverage, remedy for late delivery

10. Before you send it

- Every figure comes from a measurement or a real requirement, not a guess
- No vague words remain (“suitable”, “standard”, “high quality”)
- All requirements are in the purchase order, not only in messages
- You asked for the supplier’s reply in writing, point by point
- You have checked the supplier’s answers against the 48-point checklist

Red flags in a supplier’s reply: they avoid answering a point, they say “no problem” without figures, they say “standard” for voltage, or they refuse to put a requirement in the contract.

Want us to review your specification before you send it? We check it against all 48 verification points. WhatsApp: +2347039172842

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